Project Design Phase

Proposed Solution Template

Date	03 April 2026
Team ID	SWTID-2026-8629
Project Name	Smart Library
Maximum Marks	2 Marks

Data Flow Diagrams & Requirements – Smart Library System

Data Flow Diagrams (DFD)

A Data Flow Diagram (DFD) is a visual representation of how data flows within the Smart Library System. It shows how data enters the system, how it is processed, where it is stored, and how it is delivered as output.

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1. Functional Requirements

Functional requirements describe what the system should do.

FR-1: User Registration

The system should allow users (students/librarians) to create an account using multiple methods:

- Registration through a form (name, email, password)
- Registration using Gmail integration
- Registration using LinkedIn

FR-2: User Confirmation

After registration, the system must verify user identity:

- Email verification via link
- OTP (One-Time Password) verification through email/mobile

FR-3: User Login

The system should allow registered users to log in securely:

- Login using email and password
- Login using Gmail or LinkedIn
- Password recovery using email

FR-4: Book Management (Library Management)

The system should allow librarians/admins to manage book records:

- Add new books
- Update book details (author, category, availability)
- Delete book records

- Search and view book details

FR-5: Book Issue & Return

The system should manage book transactions:

- Issue books to users
- Accept book returns
- Track due dates and overdue books
- Calculate fines (if applicable)

FR-6: Dashboard & Reports

The system should provide insights and reports:

- View library dashboard
- Generate reports (issued books, available books, overdue books)
- Track user activity

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2. Non-Functional Requirements

Non-functional requirements define how the system performs.

NFR-1: Usability

The system should be user-friendly:

- Simple and clean interface
- Easy navigation
- Accessible for students and librarians

NFR-2: Security

The system must protect user and library data:

- Data encryption (passwords, user info)
- Secure authentication (JWT/OAuth)
- Role-based access (Admin / User)

NFR-3: Reliability

The system should work consistently:

- Minimal crashes
- Accurate book and user data processing
- Proper error handling

NFR-4: Performance

The system should respond quickly:

- Fast loading dashboards
- Efficient search and filtering

- Support multiple users at the same time

NFR-5: Availability

The system should be available anytime:

- 24/7 access
- Minimal downtime
- Backup and recovery support

NFR-6: Scalability

The system should support growth:

- Handle increasing number of users and books
- Efficient large data handling
- Flexible architecture for expansion

Conclusion

Functional requirements define what the Smart Library System does, while non-functional requirements ensure it performs efficiently, securely, and reliably. Together, they help build a user-friendly and scalable library management solution.